

Lecture 6. Sampling

When we conduct a study, we are not usually interested in the particular participants (our **sample**) – rather, we are interested in people in general (the **population**). We can never access the full population of interest in its entirety – our sample must stand in for the whole population in miniature.

If we wanted to conduct a study on the verbal skills of 5- to 6-year-old children, we would be working with a small sample of children research participants, but we would be using their performance to infer something about how children develop verbal skills in general.

Types of sampling

Random sampling: selecting members of the population at random (e.g., from a list of post codes). Random sampling is the best way of ensuring that the sample approximates the population, but true randomness is also very **difficult to achieve** in practice – for example, we could *invite* people randomly to participate in our research, but not everyone invited will agree to it. The reasons for participating in research are often *non-random* (e.g., level of education, socio-economic status), which will introduce some bias in our sampling.

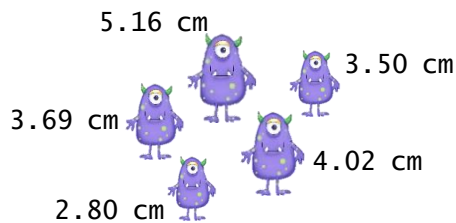
Convenience sampling: recruiting sub-populations that are easy to access (e.g., undergraduate psychology students). This is often the type of sampling used in psychological research.

Does it matter if our sampling is biased?

- **Yes** – if the non-randomness is associated with the outcome.
For example, if we only recruit children with a high socioeconomic status (SES) for a study on language development could be a problem, because we know that such children have better verbal skills and literacy on average.
- **No** – if it's unlikely to be associated with the outcome.
For example, if we only recruit people from one part of the country when looking at short term memory (no theoretical reason to think STM differs by geographical region).

Sampling Distribution

Example: We have discovered a new species of alien (population 16,870). How tall are they on average? The **population mean** will be the mean height of all 16,870 aliens, but we can't possibly measure all of them. Instead, we can get a sample of five aliens, measure their height, and find the average of that – i.e., the **sample mean**.



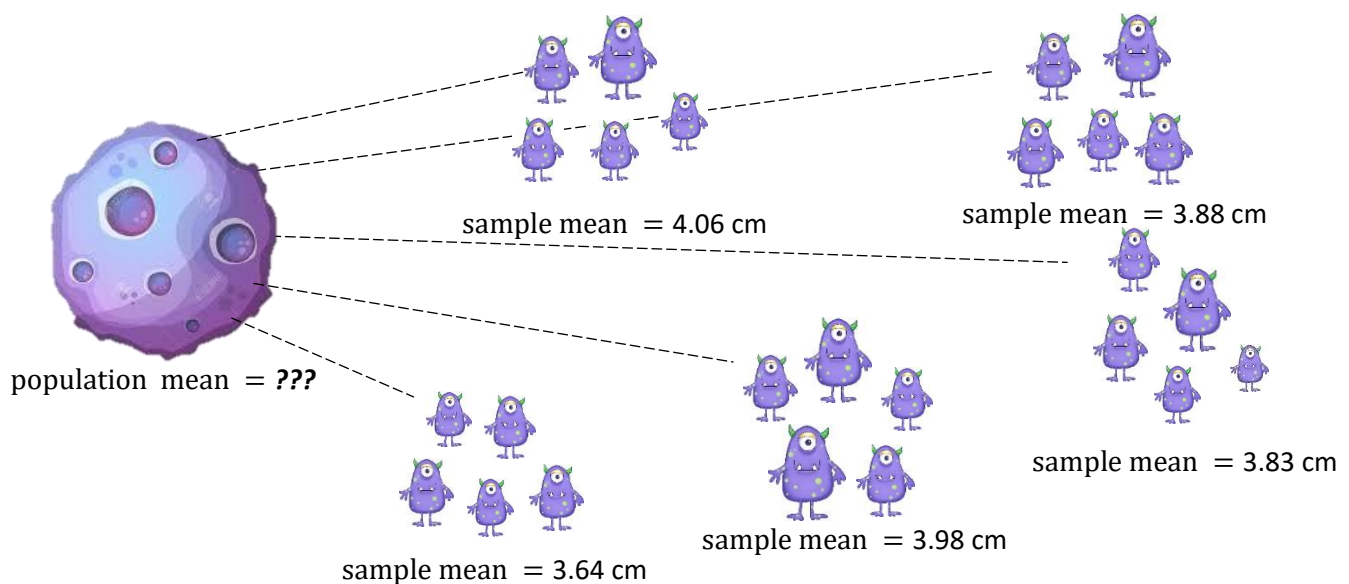
$$\text{Sample Mean} = \frac{5.16 + 3.69 + 3.50 + 4.02 + 2.80}{5} = \mathbf{3.83 \text{ cm}}$$

Notice that we still don't know what the population mean is. How likely is it that the population has the exact same mean height as our sample of five aliens? Not very likely! We call this discrepancy the **sampling error** – i.e., the difference between the means of the population and the sample.

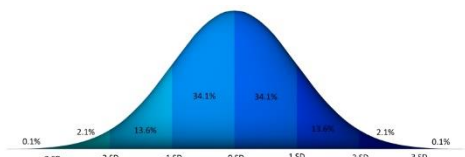
We can **reduce the sampling error** by increasing the sample size (i.e., finding more aliens whose height to measure).

But we can also quantify how close our sample mean is to the population mean. For that, we'd have to look at the **distribution** of potential samples.

There are many potential samples of 5 aliens that we can draw out of the total population of 16,870. All these samples will have slightly different means:



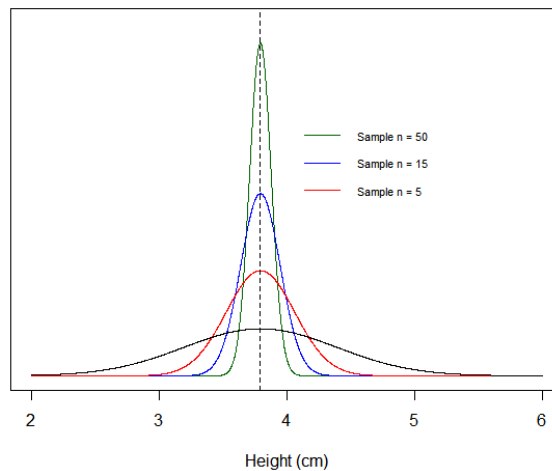
We can take all these sample means together, plot them on a histogram, and explore their distribution. They should be normally distributed – most of them should be close together, approximating the population mean, and very few of them should be far away from it. Notice that this histogram does not represent the range of heights of individual aliens – rather, this is the range of the *means* of all the different samples we've taken from the population. We call this distribution the **sampling distribution of the mean**.



The sampling distribution of the mean follows two rules in relation to the population distribution:

1. The **mean** of sample means = the population mean
2. **SD** of sample means = $\frac{SD \text{ of the population}}{\sqrt{\text{sample size } (N)}}$

The larger the sample size, the narrower the distribution of sample means. As shown on this graph, smaller samples (N=5) will have means that reflect the population mean rather poorly. Larger samples (N=50) will reflect the population more accurately, so the mean of the sample will vary less and will approximate the mean of the target population more closely.



The central limit theorem: The means of all our samples are normally distributed – even when the underlying data (e.g., heights) are not normally distributed.

Standard Error and Confidence Intervals

Any distribution of data has both a **mean** and a measure of **variability** (like a standard deviation). When we plot the distribution of all the different means of our samples, we can find not only *their* mean (i.e., the average sample mean), but their variability – which we call the standard error.

Standard error (SE) = the standard deviation of the sampling distribution of the mean. SE can be estimated from the sample standard deviation and the sample size using the following formula:

$$\text{Standard Error (SE)} = \frac{SD \text{ of the sample}}{\sqrt{\text{sample size } (N)}}$$

To continue with the above example, let's imagine that three different scientists have attempted to find the mean height of the newly discovered alien species. Each one has taken a different sample of aliens (5, 15, and 50 individuals) and calculated their mean height and its SD.

	<i>N</i>	Sample <i>M</i>	Sample <i>SD</i>
Scientist 1	5	3.83	0.866
Scientist 2	15	3.98	0.766
Scientist 3	50	3.70	0.622

Using the above formula, we can calculate the standard error of each sample (i.e., how much the sample mean deviates from the population mean):

	<i>N</i>	Sample <i>M</i>	Sample <i>SD</i>	Sample SE
Scientist 1	5	3.83	0.866	$\frac{0.866}{\sqrt{5}} = 0.387$
Scientist 2	15	3.98	0.766	$\frac{0.766}{\sqrt{15}} = 0.198$
Scientist 3	50	3.70	0.622	$\frac{0.622}{\sqrt{50}} = 0.122$

Notice how the *larger* samples produce *smaller* Standard Errors.

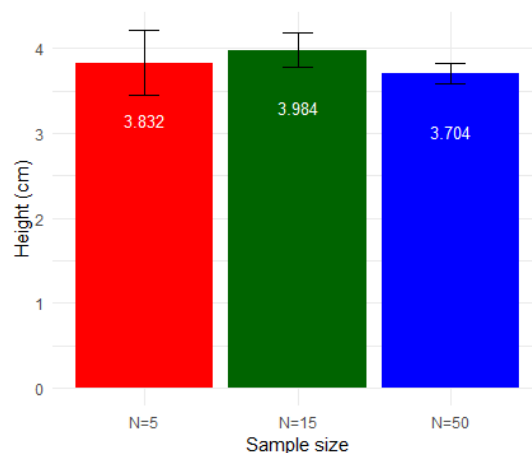
Plotting Standard Errors

We use **error bars** to plot the interval within 1 SE of the mean. There is no function to calculate SEs in R, so we must calculate it manually from the SD and sample size (e.g., $\frac{SD}{\sqrt{N}}$). For example, if we have a vector called `a` that contains a list of mean values, we can find the SE of `a` using the following code:

```
sd(a)/sqrt(length(a))
```

- **sd(a)** gives you the SD of the means;
- length(a)** gives you the length of the vector, i.e., N;
- sqrt(...)** gives you the square root of a number

Once you've calculated the SE, you can plot it as an error bar in `ggplot()` using the `geom_errorbar()` layer. Just fill in the values of "mean" and "SE":

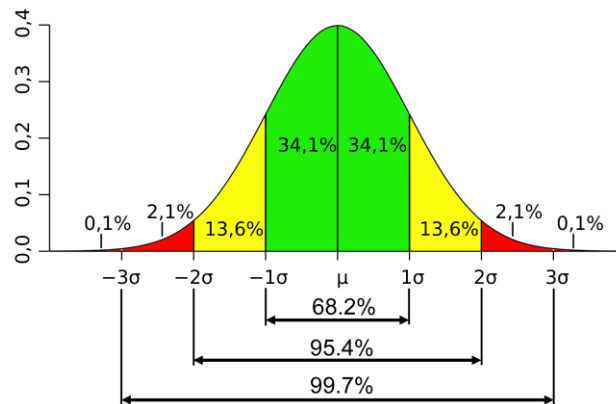


```
ggplot(data, aes(x = ..., y = ...)) +
  geom_bar() + #Add a layer for the bar chart
  geom_errorbar( #Another layer for the error bars
    aes(ymin = means - SE, #ymin sets the value of the lower bar
        ymax = means + SE), #ymax sets the value of the upper bar
    position=position_dodge(.9))
```

Interpreting the standard error (SE)

Remember this rule about normal distributions from Week 3. In any normal distribution:

- 68.2% of the data fall within ± 1 SD of the mean
- 95 % of the data fall within ± 1.96 SDs of the mean
- 95.4% of the data are within ± 2 SDs of the mean
- 99.7% of the data are within ± 3 SDs of the mean



We've used this statistical rule when exploring the distribution of data within a single sample. But it also applies to the distribution of the means of multiple samples from a population. The only difference is that the Standard Deviation (SD) measures variability within a single sample – but for our multiple samples, we have to use the Standard Error (which reflects how much any individual sample differs from the true population mean).

We can use this principle to calculate a 95% confidence interval – a range of scores that we can be 95% confident contains the true population mean.

(In other words, in practice this means that, if we sampled the same population 100 times and calculated a confidence interval ± 1.96 SEs of each sample, 95 of those CIs will contain the true population mean.)

Remember:

When you take a sample from a population, you can calculate its M and SD.

You can then use the SD and N to calculate the SE using the formula $\frac{SD}{\sqrt{N}}$.

A 95% confidence interval (CI) for the sample mean can then be constructed as $M \pm 1.96 * SE$. This interval does not guarantee that the population mean will fall within this range for any given sample, but across many samples, 95% of the intervals generated this way would contain the true population mean.

Try calculating the 95% CIs for these samples yourself:

	<i>N</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	95% CIs = $M \pm 1.96 \times SE$
Scientist 1	5	3.83	0.866	0.387	$3.83 \pm 1.96 \times 0.39$ i.e., from 3.07 to 4.59 Interpretation: we can be 95% confident that the true population mean is somewhere between 3.07 and 4.59 cm.
Scientist 2	15	3.98	0.766	0.198	

Scientist 3	50	3.70	0.622	0.122	
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Important terms:

N = sample size

M = sample Mean

SD = sample Standard Deviation (how much variability there is within our sample)

SE = Standard Error (how much the sample mean differs from the true population)

95% CI = 95% confidence interval